

MODEL EVALUATION · RESULTS ARE IN

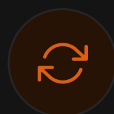
Can ZUNA clean up consumer EEG?

Testing Zyphra's EEG foundation model on two jobs — denoising what a 4-electrode headband records, and synthesizing the electrodes it doesn't have. This round: the numbers are in — spectra, decoding accuracy, and verdicts.

Gary · August 2026 · 4 subjects · Muse 2 · 60 s eyes-closed + 60 s eyes-open each

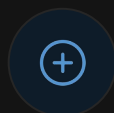
One headband, two questions

ZUNA is an open-source, 380M-parameter EEG foundation model from Zyphra. Give it a recording from a 4-electrode Muse 2 headband with parts masked out — whole channels or stretches of time — and it reconstructs the missing signal.



1 · Can it clean what we recorded?

Mask all four real channels and let ZUNA regenerate them from scratch. If the model has learned what EEG should look like, noise shouldn't survive the round trip.



2 · Can it invent what we didn't record?

Keep the four real channels untouched and ask ZUNA to synthesize four extra electrodes at the back of the head — positions the headband simply doesn't have.



● recorded by the headband (4)

○ synthesized by ZUNA (4)

top view · nose up

THE DATA

Eight minutes of labeled brain signal

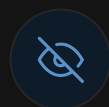
	subject	eyes closed	eyes open
4 subjects	S1	60 s label 1	60 s label 0
2 states each — EC / EO	S2	60 s label 1	60 s label 0
60 s per recording	S3	60 s label 1	60 s label 0
4 channels (Muse 2)	S4	60 s label 1	60 s label 0

Labels: eyes closed = 1 · eyes open = 0. Every subject contributes one 60-second run of each, recorded Aug 9, 2026 on a Muse 2 (dry electrodes).

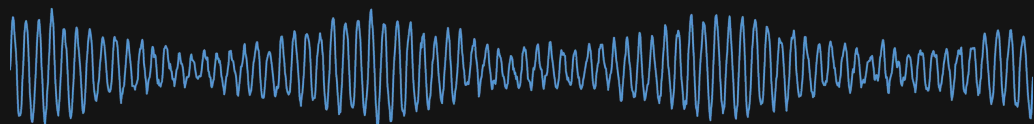
8 recordings · 8 minutes · every second labeled by what the eyes were doing

The oldest trick in EEG is our ground truth

Close your eyes and, within a second, the back of the head starts drumming at ~10 Hz. Open them and it vanishes. Hans Berger described this in 1929 — it is still the most reliable effect in all of EEG, and it happens involuntarily, every time.

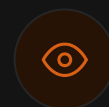


Eyes closed

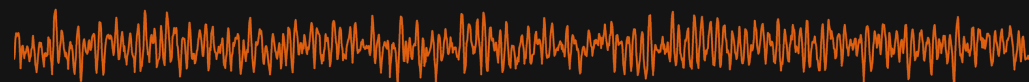


Alpha (8–12 Hz) surges, strongest over occipital cortex — the visual system idles in rhythm.

Fewer blinks, steadier eyes → less low-frequency artifact contaminating the signal.



Eyes open



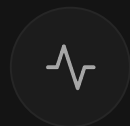
Alpha is suppressed — visual cortex is busy processing the scene, and the rhythm breaks up.

More blinks and eye movement → delta-band artifact climbs, especially on frontal electrodes.

illustrative traces — not recorded data

Why it's perfect for this eval: the difference is big, fast, and involuntary — a built-in ground truth. Processing that preserves or sharpens the EC/EO contrast is keeping real physiology; processing that kills it is inventing signal.

Three versions of every recording



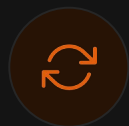
raw

the baseline

The untouched original — four channels exactly as the headband recorded them. Everything else gets judged against this.

Both eyes-closed and eyes-open runs exist in all three versions — raw feeds the “before” chart later in the deck.

no processing



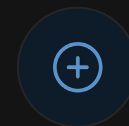
denoised

same 4 channels, regenerated

All four channels are masked in full and rebuilt from scratch. None of the original waveform survives directly — the output is ZUNA's best reconstruction of the whole signal.

Everything is masked, so full vs. hybrid outputs end up nearly identical here (see output modes).

```
zuna.reconstruct_fif(...,  
  repair_channels=[TP9, AF7, AF8, TP10])
```



upsampled

4 real + 4 synthesized

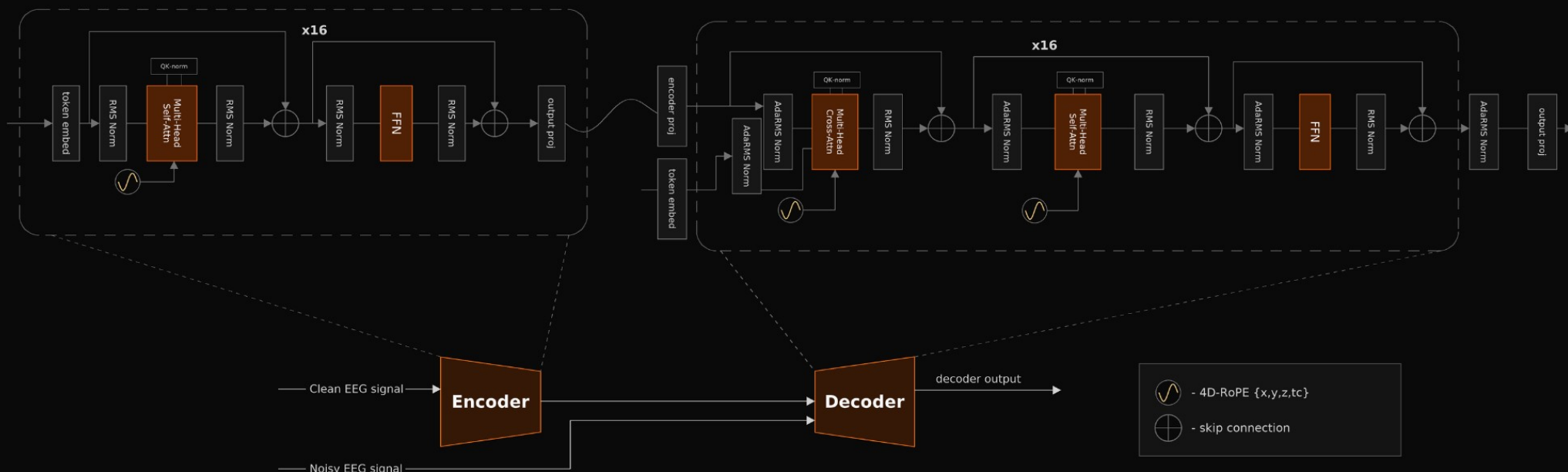
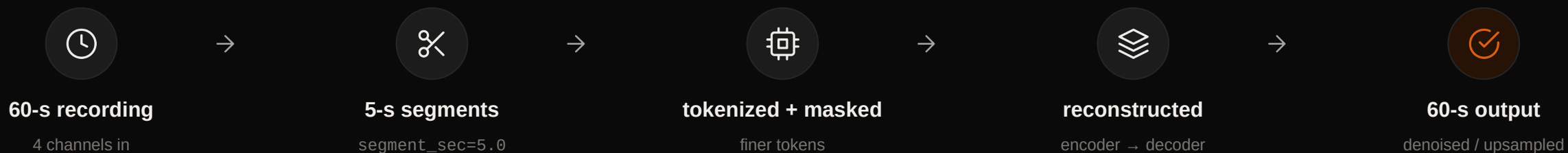
The four recorded channels are never touched. ZUNA adds four new posterior channels — O1, O2, Oz, Pz — synthesized from what the real electrodes see.

Only the new channels are masked; the real four pass through untouched.

```
zuna.reconstruct_fif(...,  
  target_channel_count=[O1, O2, Oz, Pz])
```

UNDER THE HOOD

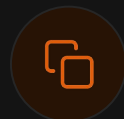
From 60 seconds of signal to a rebuilt recording



ZUNA's encoder-decoder. Masked tokens are filled in by the decoder, conditioned on everything that wasn't masked. Diagram: Zyphra.

ZYPHRA

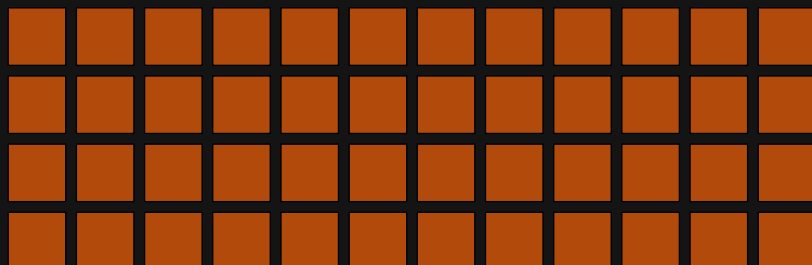
The model's repaint, or a surgical splice



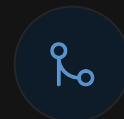
full_reconstruction/

Everything is the model

Model output for every channel and every timepoint — even ones that were never masked. A pure ZUNA view of the recording.



every cell = model output



hybrid/

Original, patched only where masked

The recording kept exactly as recorded, with model output spliced in only at the cells that were actually masked. Untouched stays untouched.

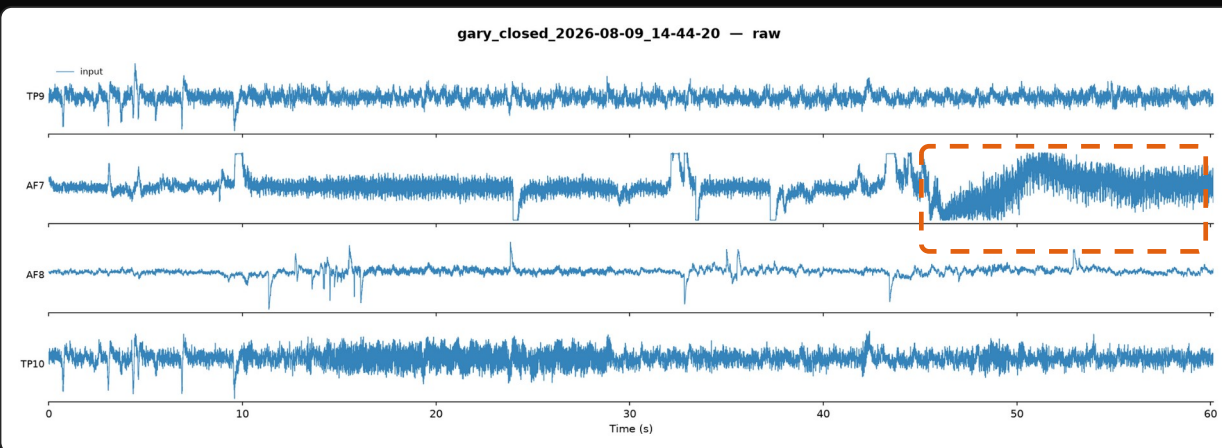


gray = original signal · orange = model, only where masked

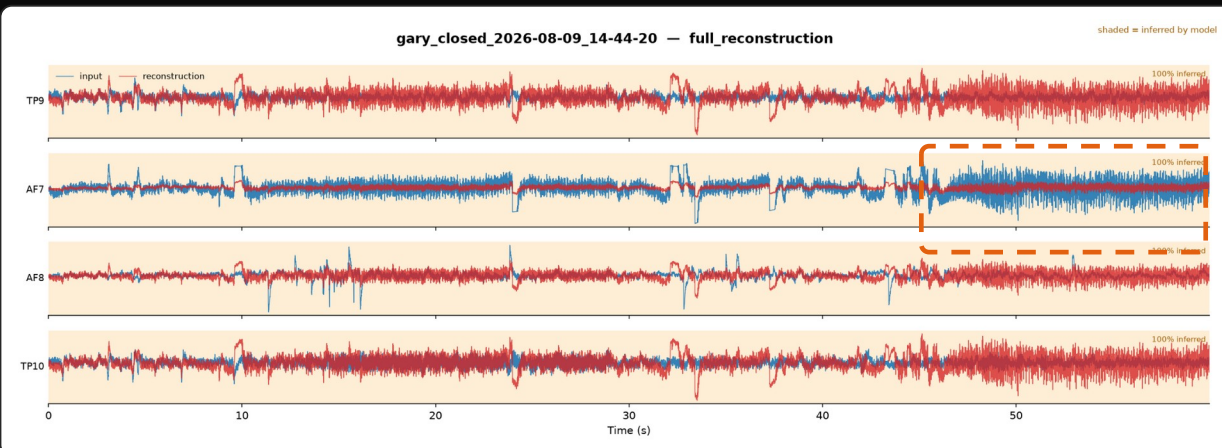
Why it matters here: in the denoised condition everything is masked, so the two modes converge — there's no untouched signal left to keep. They only really differ for upsampling.

ZUNA refuses to reproduce the artifact

RAW INPUT



FULL RECONSTRUCTION



gary_closed_2026-08-09_14-44-20 · blue = input · red = model output · shaded = inferred

ZUNA · 4-channel EEG evaluation



What to look at

Second trace: AF7, over the left eyebrow. From ~45 s the raw input saturates with high-frequency noise — muscle and motion, not brain (dashed boxes).



What ZUNA does with it

The reconstruction stays composed through the same stretch. The model won't reproduce activity it has never seen in clean EEG — the artifact simply doesn't survive.

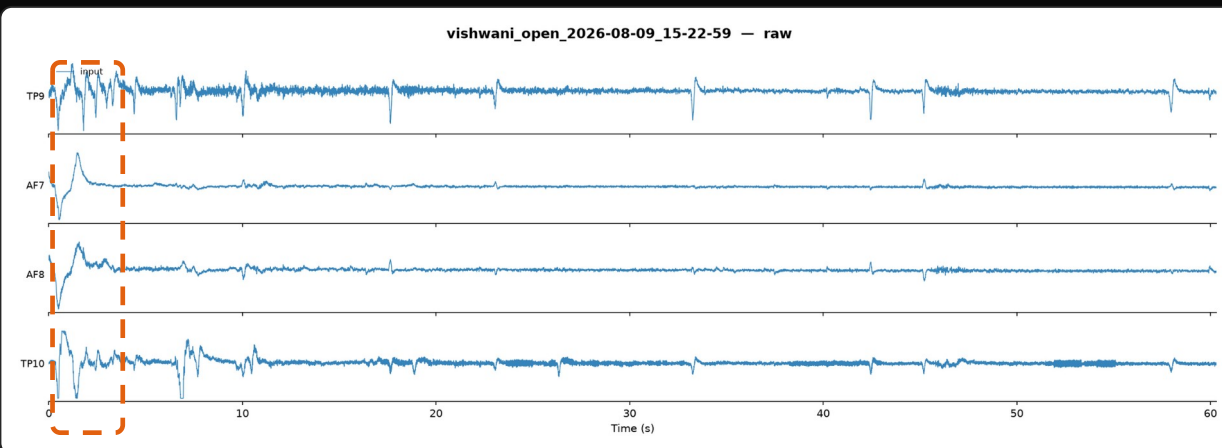


The flip side

That conservatism also softens sharp real transients. Whether cleanup costs genuine signal is exactly what the metrics on the scoreboard will decide.

When the input is clean, ZUNA leaves it alone

RAW INPUT



A cleaner recording

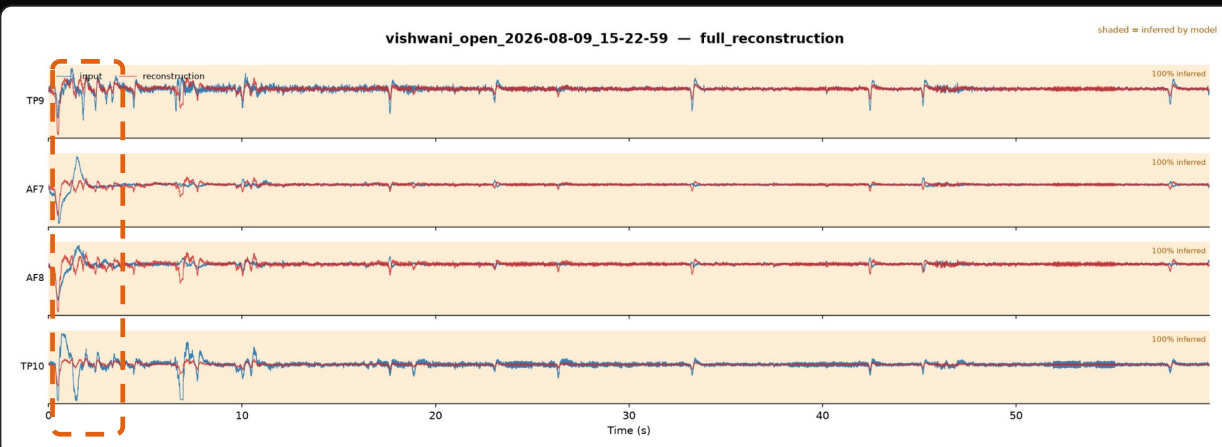
Different subject, eyes open. Apart from a settling transient in the first seconds (dashed box) and periodic blink spikes on TP9, the raw signal is well-behaved.



Reconstruction hugs the input

Red tracks blue nearly sample-for-sample across all four channels. Denoising isn't rewriting signal that doesn't need fixing.

FULL RECONSTRUCTION



Why this pair matters

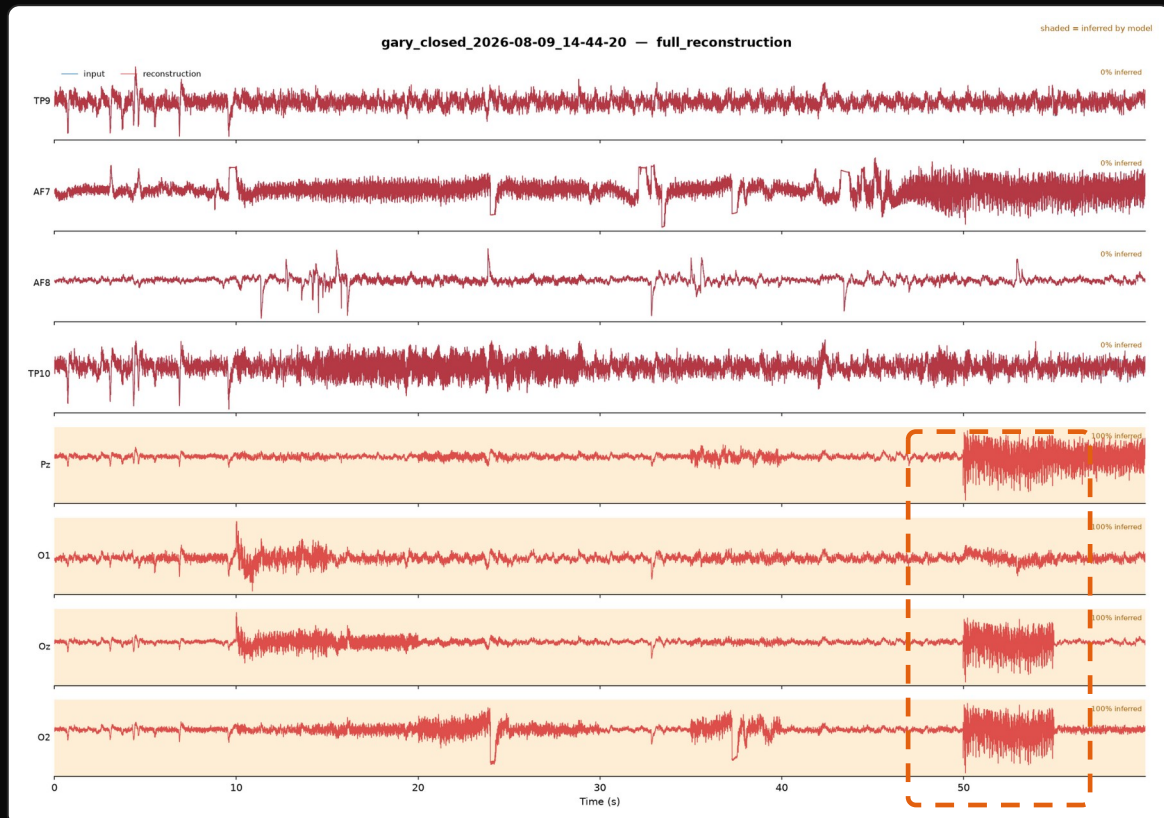
A denoiser you can trust must be aggressive on artifact and conservative on clean data. This slide and the previous one are exactly that pair.

vishwani_open_2026-08-09_15-22-59 · blue = input · red = model output · shaded = inferred

UPSAMPLING · 4 → 8 CHANNELS

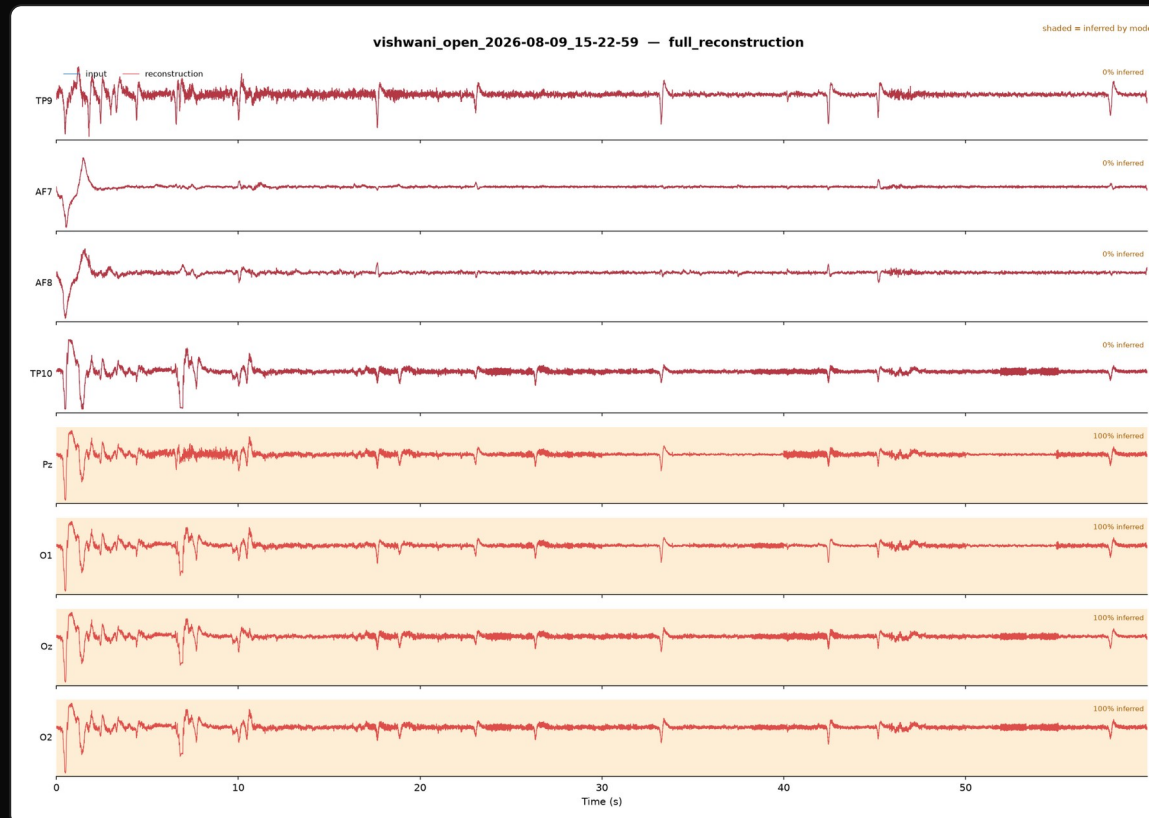
Four electrodes that were never there

EYES CLOSED · SUBJECT 1



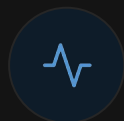
Reading it: top four rows are the real channels, passed through untouched. The shaded bottom four — Pz, O1, Oz, O2 — are pure model output: ZUNA's guess at what electrodes on the back of the head would have seen.

EYES OPEN · SUBJECT 2



Watch the eyes-closed side: structured bursts appear in the synthesized occipitals (~50 s, dashed box). If that's genuine alpha behavior, upsampling is adding physiology — not texture. The spectral check will tell.

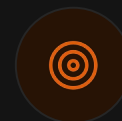
A sanity check, then the real question



spectral_check

does the alpha effect survive?

Per branch × channel: `EC_minus_EO_db` — the mean log alpha-power difference between eyes closed and eyes open, in decibels. Positive = the classic effect is present. Processing should preserve it, or reveal more of it.



classify

is the state still decodable?

Per branch: cross-validated logistic regression predicting EC vs EO from per-window alpha features → `mean_acc ± std_acc`. Folds are split by recording, so the model can't cheat by memorizing people. If accuracy holds after ZUNA touches the data, the information survived.

What passing looks like: denoised keeps (or grows) both numbers vs. raw · upsampled leaves the real channels untouched and its synthesized occipitals show the alpha effect too. Both metrics now have answers — next four slides.

How 8 recordings become one accuracy number

1

Slice

Each 60-s recording → 30 non-overlapping 2-s windows. 8 recordings → 240 windows.

2

Measure

Per window, per channel: Welch PSD → mean power in 8–13 Hz → log. One alpha number per channel.

3

Assemble

Rows = 240 windows · columns = channels (4 real; 8 when upsampled) · label = EC/EO · group = recording.

4

Split honestly

GroupKFold(5): a recording's windows never straddle train and test — no identity leakage.

5

Fit & score

Logistic regression per fold → accuracy per fold → mean ± std. One number pair per branch.

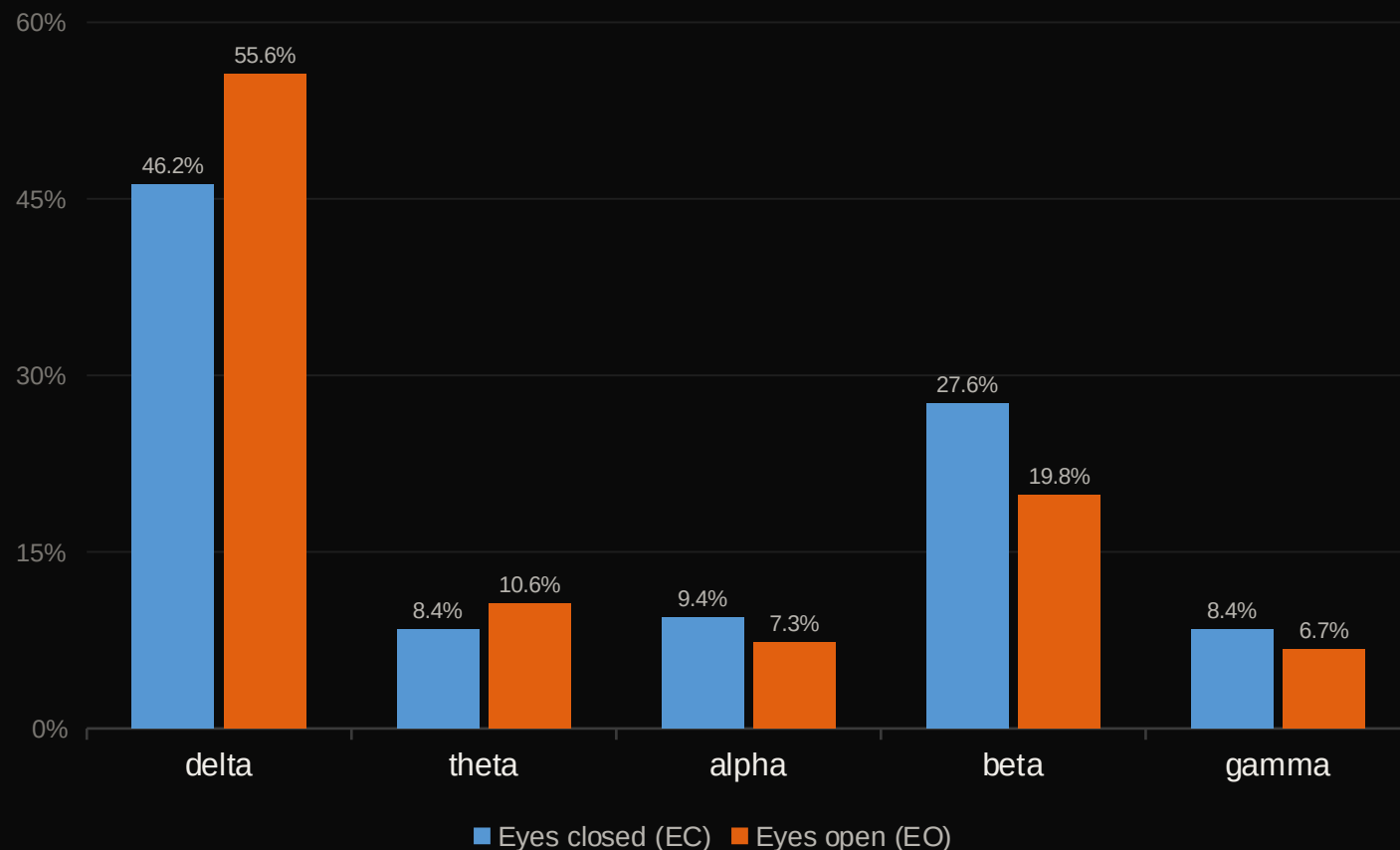


Why GroupKFold?

Neighboring windows from the same minute of EEG look almost identical. Split them randomly and the classifier recognizes the recording, not the eye state — accuracy inflates. Holding out whole recordings keeps the test honest, at the cost of brutal variance when you only have 8 of them.

8 recordings × **30** windows = **240** labeled examples per branch

The contrast is there before ZUNA touches anything



Share of total 1–45 Hz power · bands ordered slow (delta) → fast (gamma) · raw recordings, no ZUNA processing



Alpha: +2.1 pts eyes closed

9.4% vs 7.3% — the Berger effect, showing up on frontal and temporal electrodes that sit far from where alpha is strongest. Encouraging for a 4-channel headband.



Delta: higher with eyes open

55.6% vs 46.2%. Blinks live in the delta band — and you blink with your eyes open. This is artifact behaving exactly as the physiology slide predicted.



Careful — these are shares

Relative power sums to 100%, so EO's delta surplus automatically shrinks every other EO band. Some of beta's EC lead is normalization, not brain.



What the next slides test

Whether the alpha contrast survives each branch (spectral check) and whether a classifier can still find it (decoding).

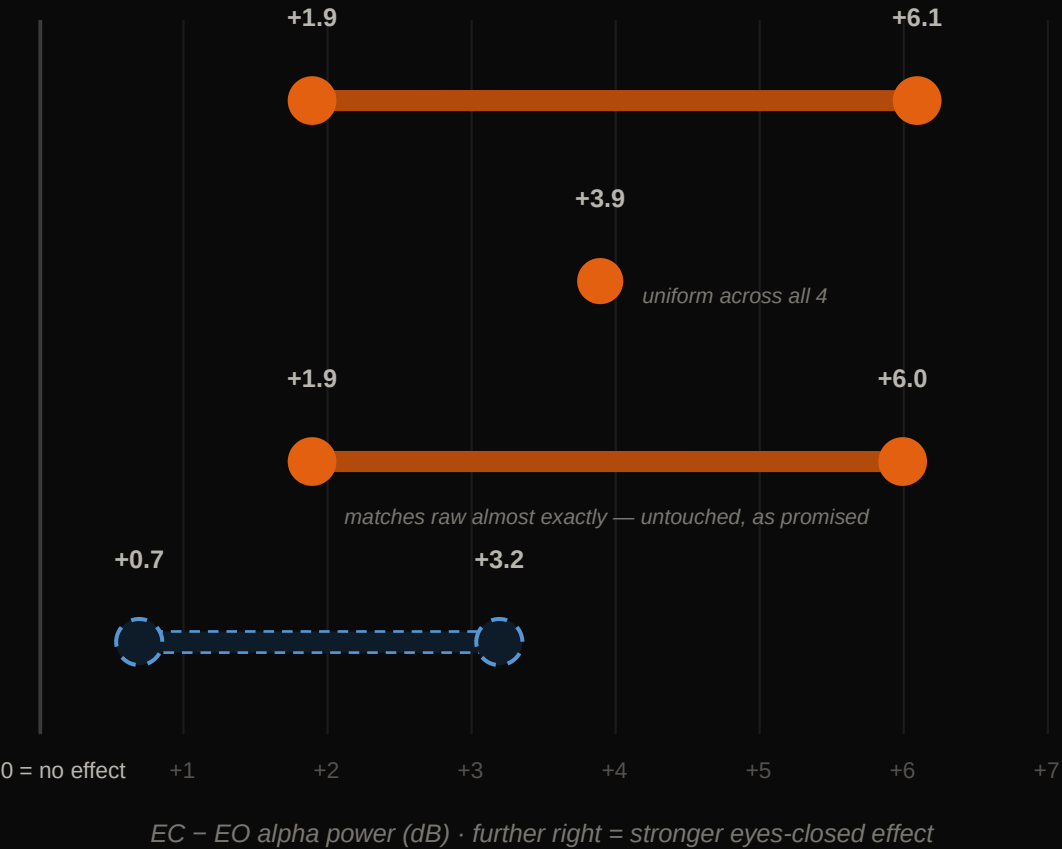
The alpha effect survives — even on invented channels

raw
4 real channels

denoised
4 real channels

upsampled
4 real channels

upsampled
O1 · O2 · Oz · Pz (synthesized)



Sanity check: PASS

Positive in every branch and channel — eyes closed always means more alpha, before and after ZUNA.



The splice is clean

Upsampled real channels return +1.9 to +6.0 dB — matching raw almost exactly. Adding channels didn't disturb the originals.



Invented channels, real physiology

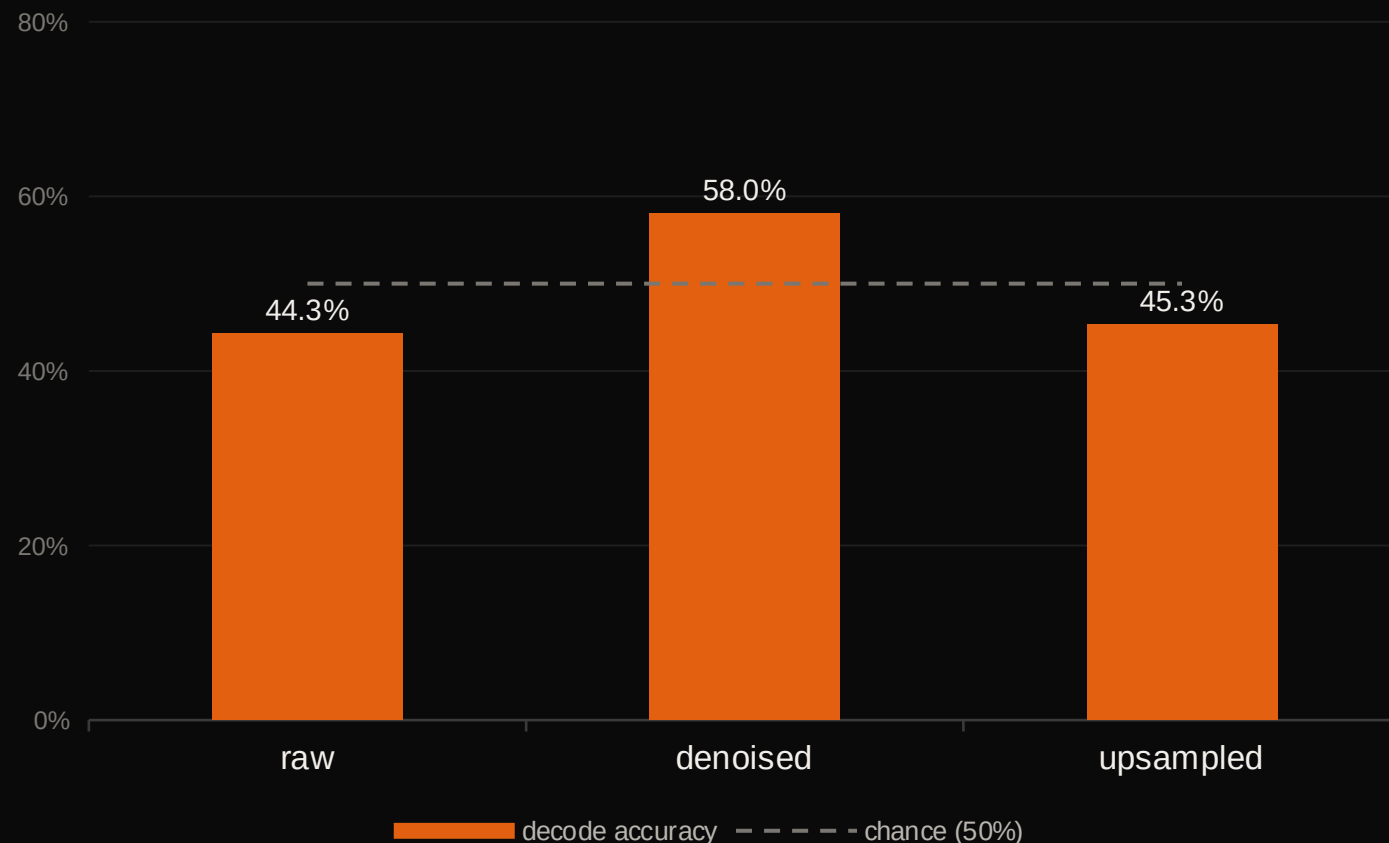
Synthesized O1/O2/Oz/Pz show the effect too — weaker than real electrodes, but present and in the right direction. The occipital question: answered.



Denoising homogenizes

All four channels land on the same +3.9 dB — per-electrode variety gets compressed. Watch this reappear in the map, two slides ahead.

The real question gets a murkier answer



Raw sits below chance

44.3% $\pm$ 27.4. With whole recordings held out and only 8 of them, folds swing wildly — read as “no reliable decoding,” not “anti-signal.”



Denoised leads at 58%

+13.7 pts over raw — directionally what denoising should do. But $\pm$ 39.8 across folds means some folds aced it and some failed completely.



Upsampled $\approx$ raw

45.3% $\pm$ 27.9. Four synthesized channels added no decodable information beyond the four they were derived from — consistent with the t-SNE overlap, next slide.



Verdict: underpowered

4 subjects $\times$ 2 recordings is tiny for recording-level cross-validation. The direction favors denoising; certainty needs more people.

n_epochs = 240 per branch by design (8×30). Source table lists 24 for upsampled — likely a dropped zero; worth confirming before sharing. · dashed line = chance

WHY DECODING IS HARD

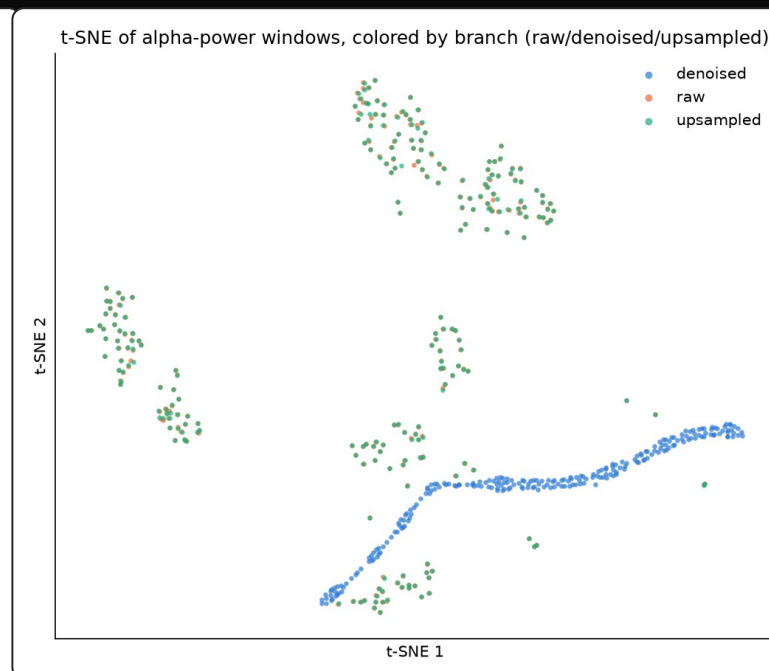
One map of every 2-second window, colored three ways

BY STATE



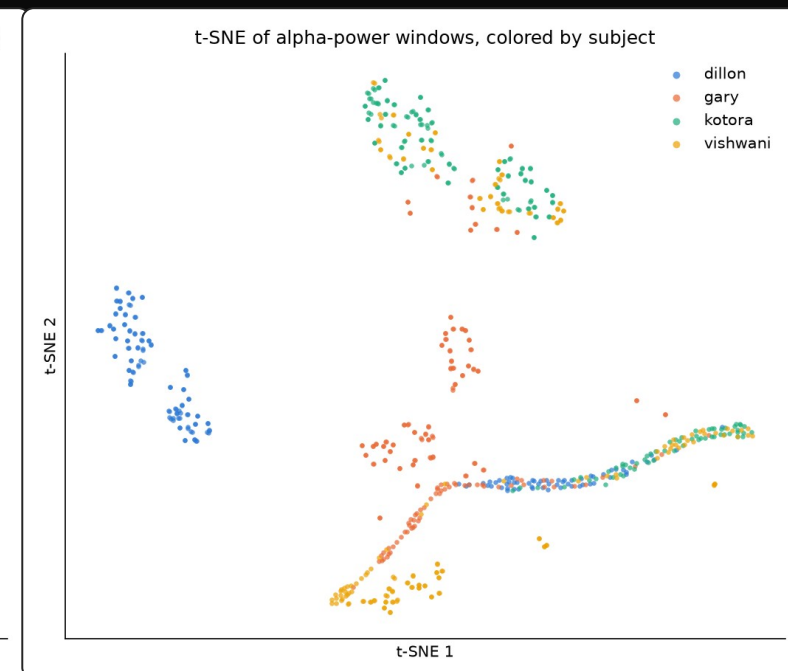
Some regions lean eyes-closed, others eyes-open — the signal exists, but the states interleave heavily. **This is what 40–60% decoding looks like.**

BY BRANCH



Denoised windows form their own connected ribbon — the channel compression from the spectral check, made visible. Raw (orange) hides beneath upsampled (green): synthesized channels barely move a window.

BY SUBJECT



The strongest structure is who you are, not what your eyes did — dillon owns an island, kotora the north. **Identity dominates alpha features; that's what makes held-out-recording decoding brutal.**

t-SNE on per-window log alpha-power features · axes are abstract — only neighborhood structure matters

Where each branch landed

branch	EC – EO alpha (dB)	decode accuracy	verdict (draft)
raw	+1.9 to +6.1	44.3% ± 27.4	baseline: effect in, decoding ~chance
denoised	+3.9 — uniform across channels	58.0% ± 39.8	effect survives, compressed · best decoder, unstable
upsampled	real: +1.9 to +6.0 (≈ raw) · synth: +0.7 to +3.2	45.3% ± 27.9	splice clean · synth carries real physiology

Bottom line (draft): ZUNA passed the physiology tests. The alpha effect survives denoising, upsampling returns the real channels untouched, and even the invented electrodes carry the eyes-closed signature. What it hasn't shown yet is a decoding win — with 4 subjects, 58% ± 40 can't be told apart from luck. The figures earn cautious optimism; the statistics ask for more people.

Sources: spectral_check and classify outputs as reported in the working deck · verdicts drafted for review

Answered this round, and what's next

Answered this round



The physiology survives ZUNA

EC – EO alpha stays positive in every branch and every channel — processing preserves the ground-truth effect.



Upsampling is honest twice over

Real channels return matching raw (+1.9 to +6.0 dB), and the synthesized occipitals show genuine — if muted — alpha behavior (+0.7 to +3.2 dB).



Denoising trades texture for clarity

Per-channel variety compresses to a uniform +3.9 dB, and denoised windows collapse onto their own manifold in the t-SNE.

Next



Power, not direction, is the gap

Denoised decodes best (58% vs 44% raw) with ± 40 -pt fold swings. More subjects — not more analysis — is the fix.



Normalize away identity

Subject clusters dominate the feature map. Per-subject normalization (or within-subject folds) could let eye state, not identity, drive the classifier.



Two loose threads

Confirm upsampled `n_epochs` (24 vs 240) in the classify log, and run hybrid vs. full on upsampled — the one condition where the modes truly differ.

Takeaways drafted by Claude from the deck's own figures and numbers — review before presenting.